

2.29 Numerical Fluid Mechanics

Spring 2015 – Lecture 9

REVIEW Lecture 8:

- **Direct Methods for solving (linear) algebraic equations**
 - Gauss Elimination
 - LU decomposition/factorization
 - Error Analysis for Linear Systems and Condition Numbers
 - Special Matrices (Tri-diagonal, banded, sparse, positive-definite, etc)

- **Iterative Methods:**

$$\mathbf{x}^{k+1} = \mathbf{B} \mathbf{x}^k + \mathbf{c} \quad k = 0, 1, 2, \dots$$

“Stationary” methods:

- **Jacobi's method**

$$\mathbf{x}^{k+1} = -\mathbf{D}^{-1}(\mathbf{L} + \mathbf{U}) \mathbf{x}^k + \mathbf{D}^{-1}\mathbf{b}$$

- **Gauss-Seidel iteration**

$$\mathbf{x}^{k+1} = -(\mathbf{D} + \mathbf{L})^{-1} \mathbf{U} \mathbf{x}^k + (\mathbf{D} + \mathbf{L})^{-1} \mathbf{b}$$

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REVIEW Lecture 8, Iterative Methods Cont'd:

– Convergence: Necessary and sufficient condition

$$\rho(\mathbf{B}) = \max_{i=1 \dots n} |\lambda_i| < 1, \quad \text{where } \lambda_i = \text{eigenvalue}(\mathbf{B}_{n \times n}) \quad (\text{ensures suffic. } \|\mathbf{B}\| < 1)$$

• Jacobi's method

• Gauss-Seidel iteration

Sufficient conditions:

- Both converge if $\mathbf{A}$ strictly diagonally dominant
- Gauss-Seidel also convergent if $\mathbf{A}$ sym. positive definite

– Stop Criteria, e.g.:

$$\begin{cases} i \leq n_{\max} \\ \|x_i - x_{i-1}\| \leq \varepsilon \\ \|r_i - r_{i-1}\| \leq \varepsilon, \text{ where } r_i = Ax_i - b \\ \|r_i\| \leq \varepsilon \end{cases}$$

– Example

– Successive Over-Relaxation Methods: (decrease $\rho(\mathbf{B})$ for faster convergence)

$$\mathbf{x}_{i+1} = (\mathbf{D} + \omega \mathbf{L})^{-1} [-\omega \mathbf{U} + (1 - \omega) \mathbf{D}] \mathbf{x}_i + \omega (\mathbf{D} + \omega \mathbf{L})^{-1} \mathbf{b}$$

“Adaptive” methods:

– Gradient Methods $\mathbf{x}_{i+1} = \mathbf{x}_i + \alpha_i \mathbf{v}_i$

• Steepest decent

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \left(\frac{\mathbf{r}_i^T \mathbf{r}_i}{\mathbf{r}_i^T \mathbf{A} \mathbf{r}_i} \right) \mathbf{r}_i$$

• Conjugate gradient

$$\begin{cases} \frac{dQ(\mathbf{x})}{d\mathbf{x}} = \mathbf{A}\mathbf{x} - \mathbf{b} = -\mathbf{r} \\ \mathbf{r}_i = \mathbf{b} - \mathbf{A}\mathbf{x}_i \text{ (residual at iteration } i) \end{cases}$$

TODAY (Lecture 9)

- End of (Linear) Algebraic Systems
 - Gradient Methods and Krylov Subspace Methods
 - Preconditioning of $Ax=b$
- **FINITE DIFFERENCES**
 - Classification of Partial Differential Equations (PDEs) and examples with finite difference discretizations
 - Error Types and Discretization Properties
 - Consistency, Truncation error, Error equation, Stability, Convergence
 - **Finite Differences based on Taylor Series Expansions**
 - Higher Order Accuracy Differences, with Example
 - Taylor Tables or Method of Undetermined Coefficients
 - Polynomial approximations
 - Newton's formulas, Lagrange/Hermite Polynomials, Compact schemes

References and Reading Assignments

- Chapter 14.2 on “Gradient Methods”, Part 8 (PT 8.1-2), Chapter 23 on “Numerical Differentiation” and Chapter 18 on “Interpolation” of “Chapra and Canale, Numerical Methods for Engineers, 2006/2010/2014.”
- Chapter 3 on “Finite Difference Methods” of “J. H. Ferziger and M. Peric, Computational Methods for Fluid Dynamics. Springer, NY, 3rd edition, 2002”
- Chapter 3 on “Finite Difference Approximations” of “H. Lomax, T. H. Pulliam, D.W. Zingg, *Fundamentals of Computational Fluid Dynamics (Scientific Computation)*. Springer, 2003”

Conjugate Gradient Method

- Derivation provided in lecture
- Check CGM_new.m

- Definition: “**A**-conjugate vectors” or “Orthogonality with respect to a matrix (metric)”:
if **A** is symmetric & positive definite,

For $i \neq j$ we say v_i, v_j are orthogonal with respect to **A**, if $v_i^T \mathbf{A} v_j = 0$

- Proposed in 1952 (Hestenes/Stiefel) so that directions v_i are generated by the orthogonalization of residuum vectors (search directions are **A**-conjugate)
 - Choose new descent direction as different as possible from old ones, within **A**-metric

- Algorithm:

$$v_0 = r_0 = b - Ax_0$$

do

$$\alpha_i = (v_i^T r_i) / (v_i^T A v_i) \quad \text{Step length}$$

$$x_{i+1} = x_i + \alpha_i v_i \quad \text{Approximate solution}$$

$$r_{i+1} = r_i - \alpha_i A v_i \quad \text{New Residual}$$

$$\beta_i = -(v_i^T A r_{i+1}) / (v_i^T A v_i) \quad \text{Step length \&}$$

$$v_{i+1} = r_{i+1} + \beta_i v_i \quad \text{new search direction}$$

until a stop criterion holds

Note: $A v_i$ = one matrix vector multiply at each iteration

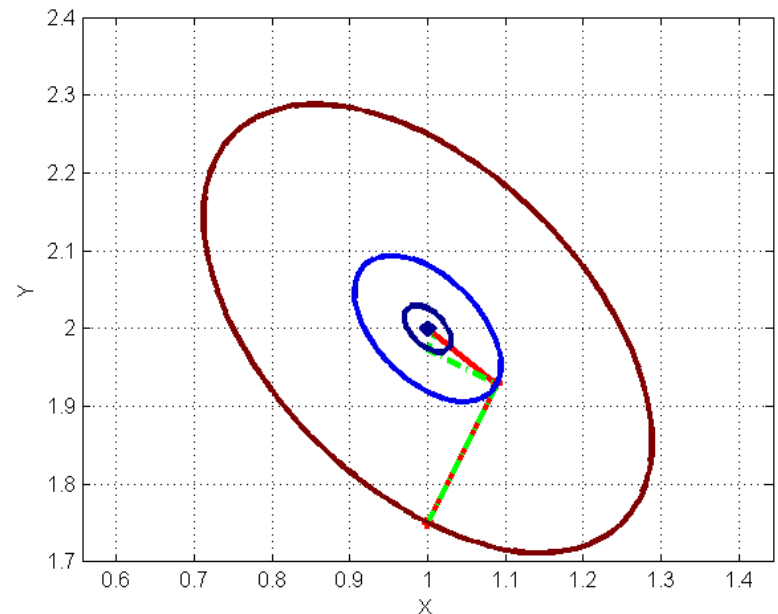


Figure indicates solution obtained using Conjugate gradient method (red) and steepest descent method (green).

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- solution with “ n ” iterations, but decent accuracy with much fewer

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$$\mathbf{Ax}=\mathbf{b}$$

-

$$\mathbf{b}, \mathbf{A} \mathbf{b}, \mathbf{A}^2 \mathbf{b}, \dots$$

-

$$n_s \leq n \quad \mathcal{K}_{n_s} = \text{span} \left\{ \mathbf{b}, \mathbf{A} \mathbf{b}, \dots, \mathbf{A}^{n_s-1} \mathbf{b} \right\}$$

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$$n_s$$

- An iteration to do this is the “Arnoldi’s iteration” which is a stabilized Gram

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$$n \quad n_s$$

-
-
- x_n are in $\mathcal{K}_n = \text{span} \left\{ \mathbf{b}, \mathbf{A} \mathbf{b}, \dots, \mathbf{A}^{n-1} \mathbf{b} \right\}$
-

- Based on the idea of projecting the “ $\mathbf{Ax}=\mathbf{b}$ problem” into the

n

$$\mathbf{Ax}=\mathbf{b}$$

$\mathbf{A}$

$$\mathbf{Ax}=\mathbf{b}$$

$$\mathbf{x}_n \in \mathcal{K}_n$$

$$\mathbf{Ax}_n - \mathbf{b}$$

Preconditioning of $\mathbf{A} \mathbf{x} = \mathbf{b}$

conditioner approximately solves $\mathbf{A} \mathbf{x} = \mathbf{b}$.

Pre-multiply by the inverse of a non-singular matrix $\mathbf{M}$, and solve instead:

$$\mathbf{M}^{-1}\mathbf{A} \mathbf{x} = \mathbf{M}^{-1} \mathbf{b} \quad \text{or} \quad \mathbf{A} \mathbf{M}^{-1} (\mathbf{M} \mathbf{x}) = \mathbf{b}$$

- Convergence properties based on $\mathbf{M}^{-1}\mathbf{A}$ or $\mathbf{A} \mathbf{M}^{-1}$ instead of $\mathbf{A}$!
- Can accelerate subsequent application of iterative schemes
- Can improve conditioning of subsequent use of non-iterative schemes: GE, LU, etc

- Jacobi preconditioning:
 - Apply Jacobi a few steps, usually not efficient
- Other iterative methods (Gauss-Seidel, SOR, SSOR, etc):
 - Usually better, sometimes applied only once
- Incomplete factorization (incomplete LU) or incomplete Cholesky
 - LU or Cholesky, but avoiding fill-in of already null elements in $\mathbf{A}$
- Coarse-Grid Approximations and Multigrid Methods:
 - Solve $\mathbf{A} \mathbf{x} = \mathbf{b}$ on a coarse grid (or successions of coarse grids)
 - Interpolate back to finer grid(s)

Example of Convergence Studies for Linear Solvers

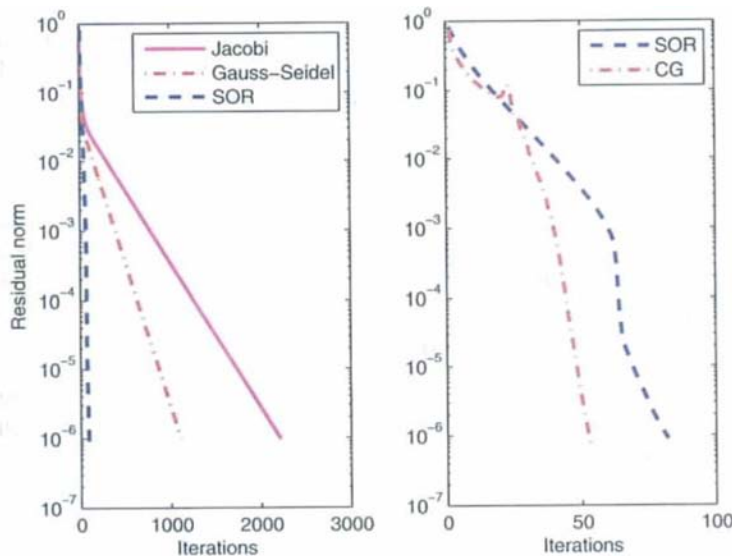


Fig 7.5: Example 7.10, with system of size 961x961: convergence behavior of various iterative schemes for the discretized Poisson equation.

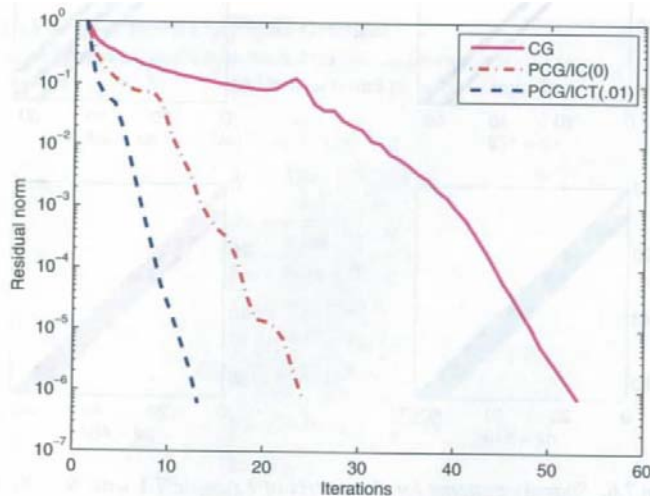


Fig 7.7: Iteration progress for CG, PCG with the IC(0) preconditioner and PCG with the IC preconditioner using drop tolerance $\text{tol}=0.01$

IC(0): is incomplete Cholesky factorization. This is Cholesky as we have seen it, but a non-zero entry in the factorization is generated only if A was not zero there to begin with.

IC: same, but non-zero entry generated if it is $\geq \text{tol}$

PCG: Preconditioned conjugate gradient

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Ascher and Greif (SIAM-2011)

Review of/Summary for Iterative Methods

Table removed due to copyright restrictions. Useful reference tables for this material:
Tables PT3.2 and PT3.3 in Chapra, S., and R. Canale. *Numerical Methods for Engineers*.
6th ed. McGraw-Hill Higher Education, 2009. ISBN: 9780073401065.

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FINITE DIFFERENCES - Outline

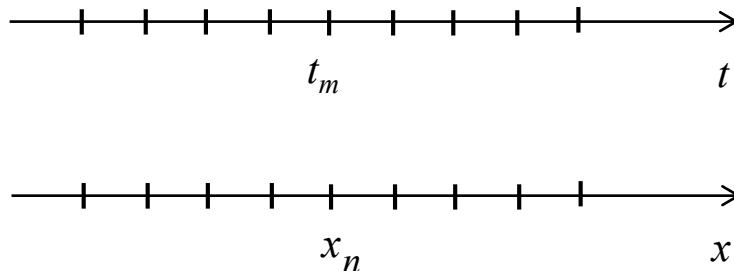
- **Classification of Partial Differential Equations (PDEs) and examples with finite difference discretizations**
 - Elliptic PDEs
 - Parabolic PDEs
 - Hyperbolic PDEs
- **Error Types and Discretization Properties**
 - Consistency, Truncation error, Error equation, Stability, Convergence
- **Finite Differences based on Taylor Series Expansions**
- **Polynomial approximations**
 - Equally spaced differences
 - Richardson extrapolation (or uniformly reduced spacing)
 - Iterative improvements using Roomberg's algorithm
 - Lagrange polynomial and un-equally spaced differences
 - Compact Difference schemes

Continuum Model

$$\frac{\partial w}{\partial t} + c \frac{\partial w}{\partial x} = 0$$

Sommerfeld Wave Equation (c = wave speed).
This radiation condition is sometimes used at open boundaries of ocean models.

Discrete Model



$$t_m = t_0 + m \Delta t, \quad m = 0, 1, \dots, M-1$$

$$x_n = x_0 + n \Delta x, \quad n = 0, 1, \dots, N-1$$

$$\frac{\partial w}{\partial t} \simeq \frac{\Delta w}{\Delta t}, \quad \frac{\partial w}{\partial x} \simeq \frac{\Delta w}{\Delta x}$$

p parameters, e.g. variable c

Differential Equation

$$L(p, w, x, t) = 0$$

“Differentiation”
“Integration”

Difference Equation

$$L_{mn}(p_{mn}, w_{mn}, x_n, t_m) = 0$$

System of Equations

$$\sum_{j=0}^{N-1} F_i(w_j) = B_i$$

Linear System of Equations

$$\sum_{j=0}^{N-1} A_{ij} w_j = B_i$$

“Solving linear equations”

Eigenvalue Problems

$$\bar{\bar{\mathbf{A}}} \mathbf{u} = \lambda \mathbf{u} \Leftrightarrow (\bar{\bar{\mathbf{A}}} - \lambda \bar{\bar{\mathbf{I}}}) \mathbf{u} = 0$$

Non-trivial Solutions

$$\det(\bar{\bar{\mathbf{A}}} - \lambda \bar{\bar{\mathbf{I}}}) = 0$$

“Root finding”

Consistency/Accuracy and Stability => Convergence
(Lax equivalence theorem for well-posed linear problems)

Classification of Partial Differential Equations

(2D case, 2nd order PDE)

Quasi-linear PDE for $\phi(x, y)$

$$A\phi_{xx} + B\phi_{xy} + C\phi_{yy} = F(x, y, \phi, \phi_x, \phi_y)$$

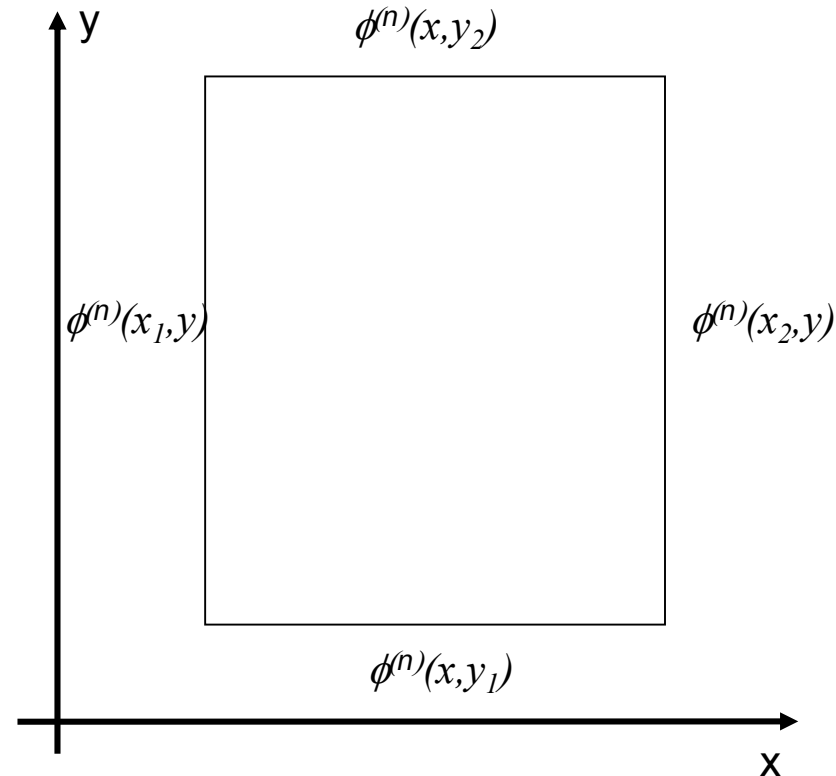
A, B and C Constants

$$B^2 - 4AC > 0 \quad \text{Hyperbolic}$$

$$B^2 - 4AC = 0 \quad \text{Parabolic}$$

$$B^2 - 4AC < 0 \quad \text{Elliptic}$$

(Only valid for two independent variables: x, y)



- In general: A, B and C are function of: $x, y, \phi, \phi_x, \phi_y$
- Equations may change of type from point to point if A, B and C vary with x, y , etc.
- Navier-Stokes, incomp., const. viscosity:
$$\frac{D\mathbf{u}}{Dt} = \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{g}$$

Classification of Partial Differential Equations

(2D case, 2nd order PDE)

Meaning of Hyperbolic, Parabolic and Elliptic

- The general 2nd order PDE in 2D:

$$A\phi_{xx} + B\phi_{xy} + C\phi_{yy} = F$$

is analogous to the equation for a conic section:

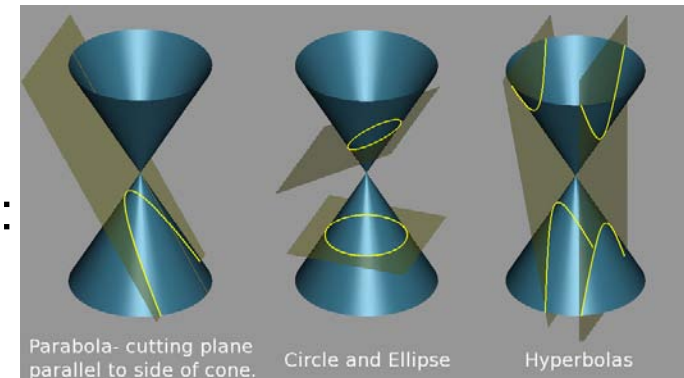
$$Ax^2 + Bxy + Cy^2 = F$$

- Conic section:

- Is the intersection of a right circular cone and a plane, which generates a group of plane curves, including the circle, ellipse, hyperbola, and parabola
- One characterizes the type of conic sections using the discriminant $B^2 - 4AC$

- PDE:

- $B^2 - 4AC > 0$ (Hyperbolic)
- $B^2 - 4AC = 0$ (Parabolic)
- $B^2 - 4AC < 0$ (Elliptic)



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Examples

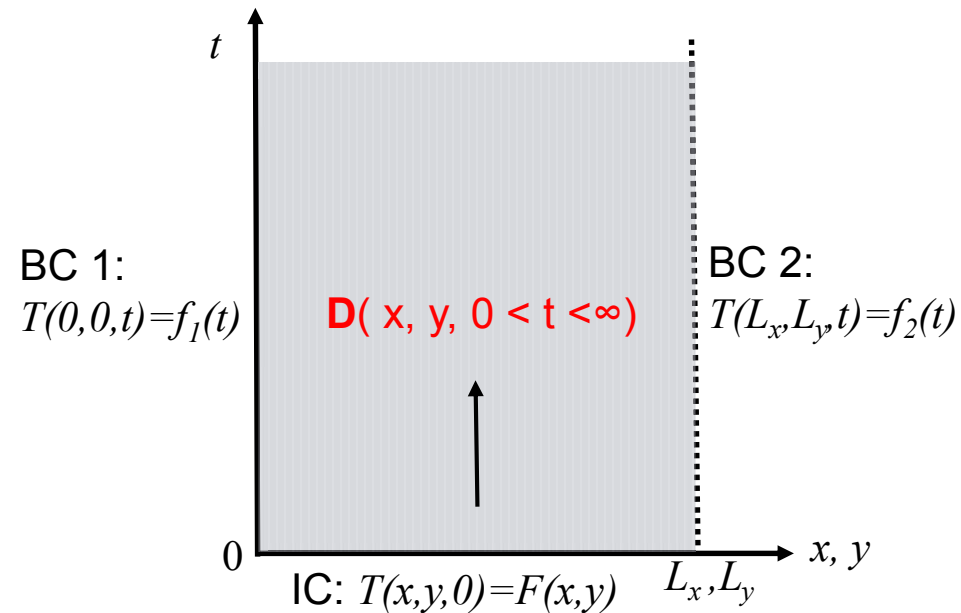
$$\frac{\partial T}{\partial t} = \frac{\kappa}{\rho c} \nabla^2 T + f, \quad (\alpha = \frac{\kappa}{\rho c})$$

Heat conduction equation,
forced or not (dominant in 1D)

$$\frac{\partial \mathbf{u}}{\partial t} = \nu \nabla^2 \mathbf{u} + \mathbf{g}$$

Unsteady, diffusive, small amplitude flows
or perturbations (e.g. Stokes Flow)

- Usually smooth solutions (“diffusion effect” present)
- “Propagation” problems
- Domain of dependence of solution is domain **D** (x, y , and $0 < t < \infty$):
- Finite Differences/Volumes, Finite Elements



Partial Differential Equations

Parabolic PDE - Example

Heat Conduction Equation

$$\kappa T_{xx}(x,t) = \rho c T_t(x,t), 0 < x < L, 0 < t < \infty$$

Initial Condition

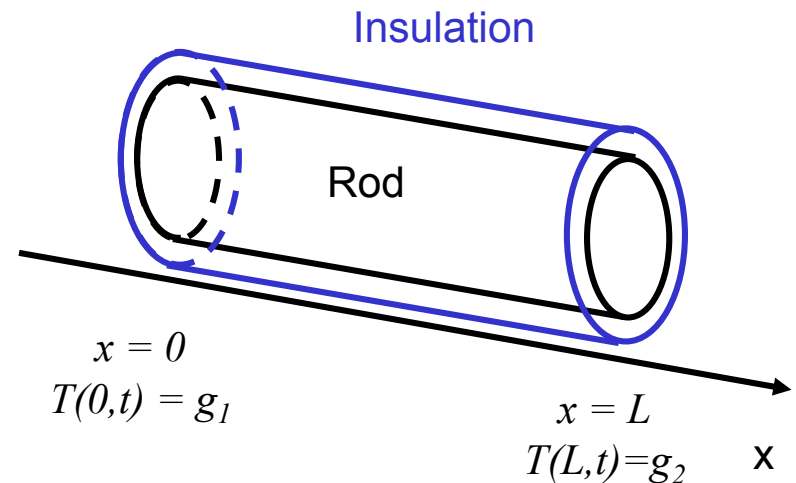
$$T(x,0) = f(x), 0 \leq x \leq L$$

Boundary Conditions

$$T(0,t) = g_1, 0 < t < \infty$$

$$T(L,t) = g_2, 0 < t < \infty$$

κ Thermal conductivity
 c Specific heat capacity
 ρ Density
 T Temperature



IVP in one dimension (t), BVP in the other (x)
Time Marching, Explicit or Implicit Schemes

IVP: Initial Value Problem
BVP: Boundary Value Problem

Partial Differential Equations

Parabolic PDE - Example

Heat Conduction Equation

$$T_t(x, t) = \alpha T_{xx}(x, t), 0 < x < L, 0 < t < \infty$$

$$\alpha = \frac{\kappa}{\rho c}$$

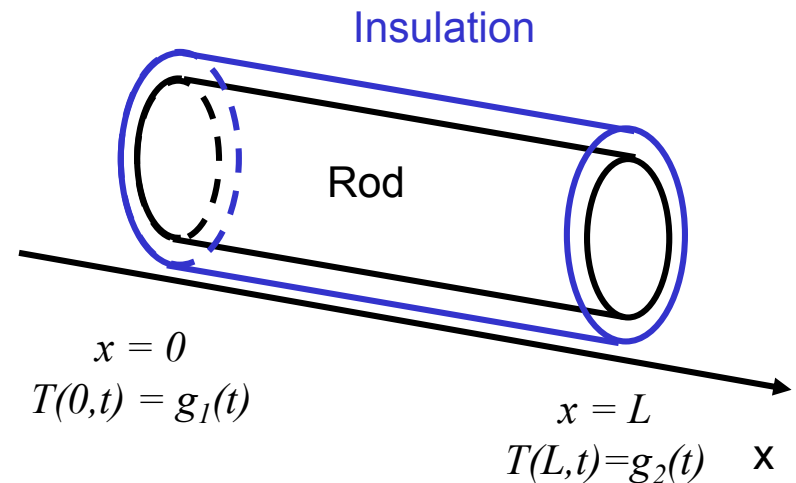
Initial Condition

$$T(x, 0) = f(x), 0 \leq x \leq L$$

Boundary Conditions

$$T(0, t) = g_1(t), 0 < t < \infty$$

$$T(L, t) = g_2(t), 0 < t < \infty$$



Partial Differential Equations

Parabolic PDE - Example

Equidistant Sampling

$$h = L/n$$

$$k = T/m$$

Discretization

$$x_i = (i-1)h, \quad i = 2, \dots, n-1$$

$$t_j = (j-1)k, \quad j = 1, \dots, m$$

Forward (Euler) Finite Difference in time

$$T_t(x, t) = \frac{T(x_i, t_{j+1}) - T(x_i, t_j)}{k} + O(k)$$

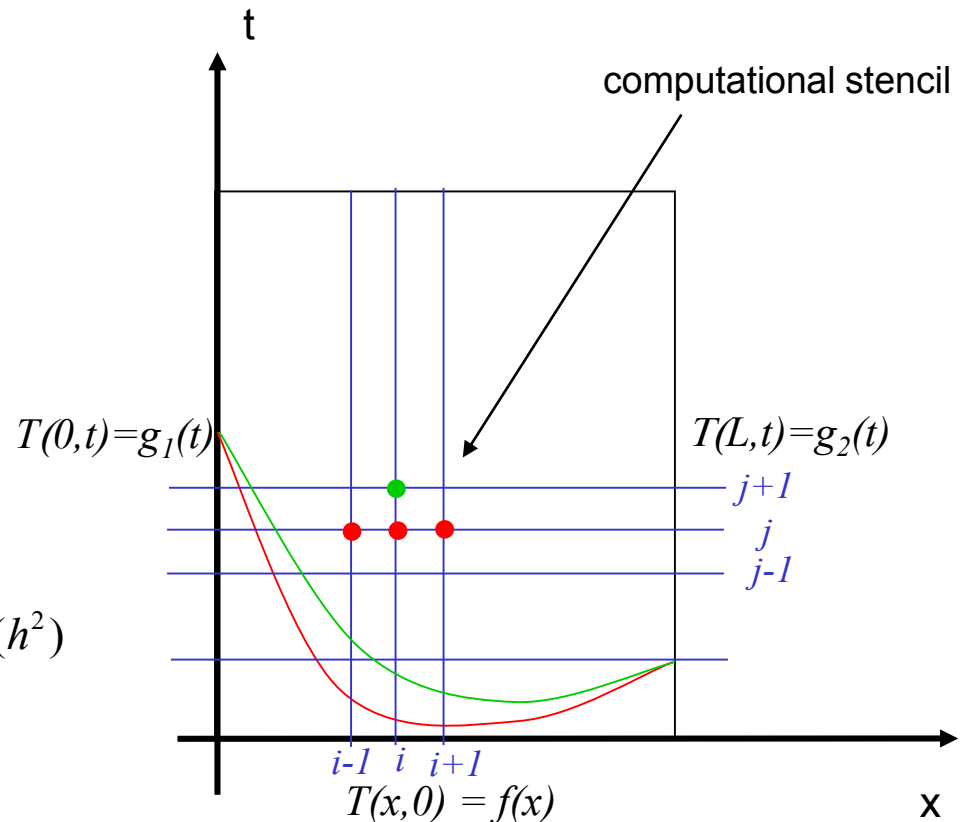
Centered Finite Difference in space

$$T_{xx}(x, t) = \frac{T(x_{i-1}, t_j) - 2T(x_i, t_j) + T(x_{i+1}, t_j))}{h^2} + O(h^2)$$

$$T_{i,j} = T(x_i, t_j)$$

Finite Difference Equation

$$\frac{T_{i,j+1} - T_{i,j}}{k} = \alpha \frac{T_{i-1,j} - 2T_{i,j} + T_{i+1,j}}{h^2}$$



Partial Differential Equations

ELLIPTIC: $B^2 - 4AC < 0$

Quasi-linear PDE for $\phi(x, y)$

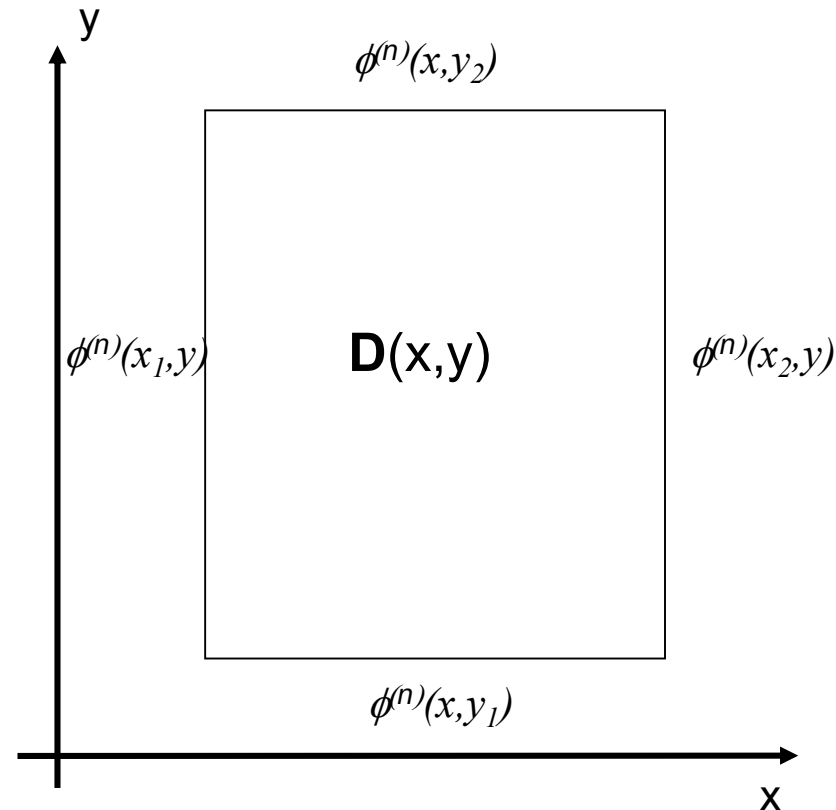
$$A\phi_{xx} + B\phi_{xy} + C\phi_{yy} = F(x, y, \phi, \phi_x, \phi_y)$$

A, B and C Constants

$$B^2 - 4AC > 0 \quad \text{Hyperbolic}$$

$$B^2 - 4AC = 0 \quad \text{Parabolic}$$

$$B^2 - 4AC < 0 \quad \text{Elliptic}$$



Partial Differential Equations

Elliptic PDE

Laplace Operator

$$\nabla^2 \equiv u_{xx} + u_{yy}$$

Examples:

$$\nabla^2 \phi = 0$$

Laplace Equation – Potential Flow

$$\nabla^2 \phi = g(x, y)$$

Poisson Equation

- Potential Flow with sources
- Steady heat conduction in plate + source

$$\nabla^2 u + f(x, y)u = 0$$

Helmholtz equation – Vibration of plates

$$\mathbf{U} \cdot \nabla \mathbf{u} = \nu \nabla^2 \mathbf{u}$$

Steady Convection-Diffusion

- Smooth solutions (“diffusion effect”)
- Very often, steady state problems
- Domain of dependence of u is the full domain $\mathbf{D}(x, y) \Rightarrow$ “global” solutions
- Finite differ./volumes/elements, boundary integral methods (Panel methods)

Partial Differential Equations

Elliptic PDE - Example

$$0 \leq x \leq a, \quad 0 \leq y \leq b;$$

Equidistant Sampling

$$h = a/(n-1)$$

$$h = b/(m-1)$$

Discretization

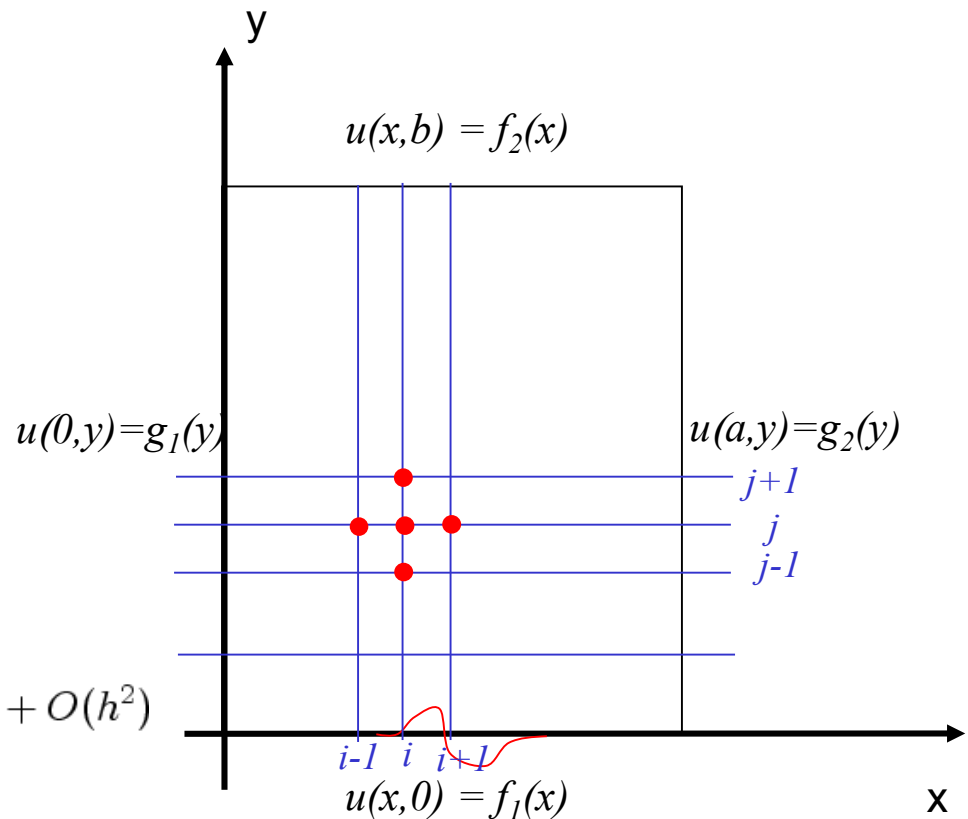
$$x_i = (i-1)h, \quad i = 1, \dots, n$$

$$y_j = (j-1)h, \quad j = 1, \dots, m$$

Finite Differences

$$u_{xx}(x, t) = \frac{u(x_{i-1}, y_j) - 2u(x_i, y_j) + u(x_{i+1}, y_j)}{h^2} + O(h^2)$$

$$u_{yy}(x, t) = \frac{u(x_i, y_{j-1}) - 2u(x_i, y_j) + u(x_i, y_{j+1}))}{h^2} + O(h^2)$$



Dirichlet BC

Partial Differential Equations

Elliptic PDE - Example

Discretized Laplace Equation

$$\nabla^2 u = \frac{u(x_{i-1}, y_j) + u(x_i, y_{j-1}) - 4u(x_i, y_j) + u(x_{i+1}, y_j) + u(x_i, y_{j+1})}{h^2} = 0$$

$$u_{i,j} = u(x_i, y_j)$$

Finite Difference Scheme

$$u_{i+1,j} + u_{i-1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{i,j} = 0$$

Boundary Conditions

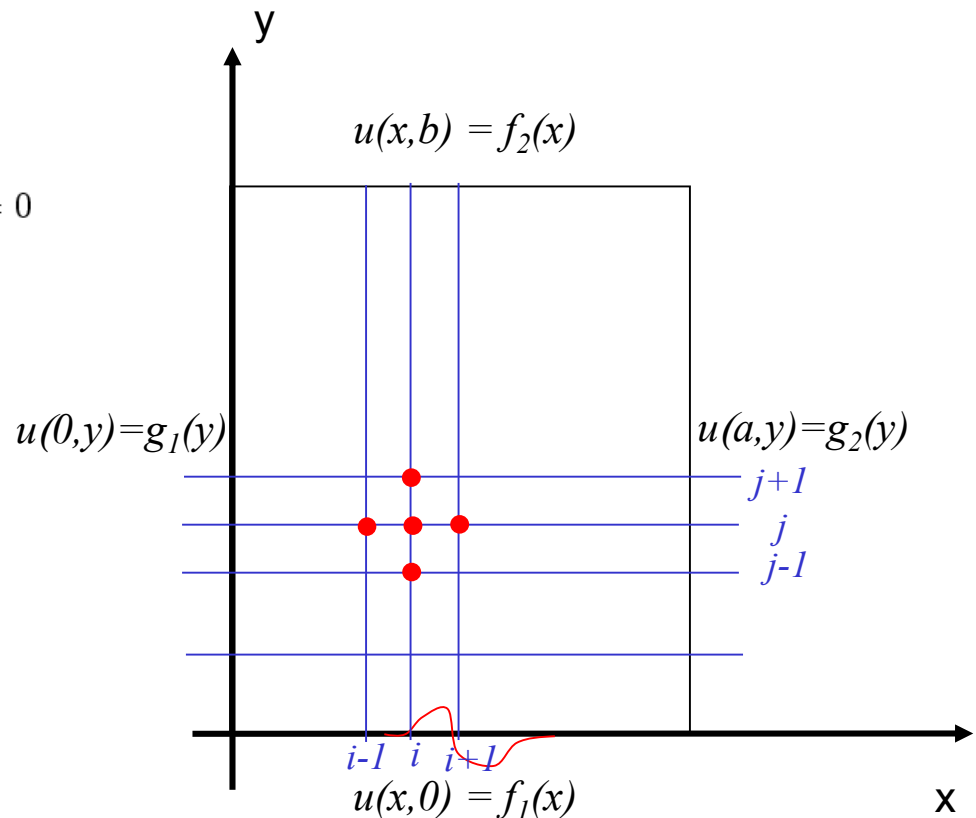
$$u(x_1, y_j) = u_{1,j}, \quad 2 \leq j \leq m-1$$

$$u(x_n, y_j) = u_{n,j}, \quad 2 \leq j \leq m-1$$

$$u(x_i, y_1) = u_{i,1}, \quad 2 \leq i \leq n-1$$

$$u(x_i, y_n) = u_{i,n}, \quad 2 \leq i \leq n-1$$

Global Solution Required



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Spring 2015

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